Quantum Mechanics

Leonard Susskind

Core track, 2012 Winter Theoretical Minimum Alt Title
Lecture notes organized by LazyingArt LLC with Video2Book

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Chapter 1

Lecture 1: The Theoretical Minimum

These notes follow the first quantum-mechanics lecture as a guided unfolding rather than as a retrospective textbook summary. We begin where the lecture begins, with classical mechanics as the theory of states and deterministic updating. We then pass through a warning about visualization, into the simplest quantum experiment, and only afterward into the abstract mathematics of complex vector spaces. The lesson of the lecture is not yet the full formalism of quantum mechanics. It is the more primitive fact that the quantum notion of state is already unlike the classical one, and that the mathematics must be prepared before the physics can be represented properly.

1.1 From Classical Mechanics to Quantum Abstraction

Last quarter's subject was classical mechanics, and the lecture begins by rebuilding its basic structure from the bottom up. First comes the notion of a *system*: a closed, isolated thing whose history we want to predict. Next comes the notion of a *state*: the information that must be specified at an instant in order to predict the future. Finally come the *equations of motion*: the rules that update the state from one time to the next.

For a toy system, the structure is almost embarrassingly simple. If the system is an abstract coin, one possible law of motion is that nothing happens:

$$H \longmapsto H, \qquad\qquad T \longmapsto T. \tag{1.1}$$

Another possible law is alternation:

$$H \longmapsto T, \qquad\qquad T \longmapsto H. \tag{1.2}$$

These examples are trivial, but they make the classical pattern explicit. Classical mechanics is a theory of updating:

$$s(t) \longmapsto s(t + \Delta t), \tag{1.3}$$

and, once the state $s(t)$ is known, the update is deterministic.

As we move from discrete examples to continuous motion, the state space becomes larger and more abstract, but the structure is unchanged. A particle moving on a line no longer has only two states, or six, or any finite number. Its state must be labeled by continuous data. Still, the theory remains predictive: give the state at one time, and the equations tell us what the state will be at the next.

At this point the lecture pauses for a warning that is not ornamental. Once physics moves beyond the range of ordinary experience, visualization becomes unreliable. We cannot picture the motion of an electron in any literal everyday sense. We cannot visualize four-dimensional spacetime, much less higher-dimensional spaces. Susskind's example is deliberately sharp: we cannot even truly visualize a two-dimensional curved space without sneaking in a three-dimensional embedding. If we try to understand quantum mechanics by picturing it as a slightly funny sort of classical mechanics, we will almost certainly get it wrong.

The cure is not better pictures. The cure is abstraction. We learn to manipulate the mathematics faithfully, and only afterward do we develop a more appropriate kind of intuition. This methodological warning is the hinge of the lecture. Without it, the later move to vector spaces would feel unmotivated; with it, the mathematics arrives exactly when it is needed.

1.2 Classical State Spaces and Classical Logic

Armed with that warning, the lecture returns to a classical question: what kind of thing is a space of states? For classical systems, the answer is that the state space is a set.

For an abstract coin,

$$\mathcal{S}_{\text{coin}} = \{H, T\}. \tag{1.4}$$

For an abstract die,

$$\mathcal{S}_{\text{die}} = \{1, 2, 3, 4, 5, 6\}. \tag{1.5}$$

For a point particle moving on a line, the state consists of position together with momentum; the appropriate classical state space is phase space,

$$\mathcal{S}_{\text{particle}} = \{(q, p) \mid q, p \in \mathbb{R}\}. \tag{1.6}$$

In the first two cases the state space is finite; in the third it is continuously infinite. But in every case it is still a set of possible states.

Once the state space is a set, classical logic falls into place almost automatically. A proposition is represented by a subset of the state space, and logical operations become set-theoretic operations. If P and Q are propositions, then

$$P \wedge Q \leftrightarrow P \cap Q, \qquad P \vee Q \leftrightarrow P \cup Q, \tag{1.7}$$

with $\vee$ understood as the inclusive "or."

The lecture uses the die to make this completely concrete. Let

$$P_{\text{odd}} = \{1, 3, 5\}, \tag{1.8}$$

$$Q_{\leq 3} = \{1, 2, 3\}. \tag{1.9}$$

Then

$$P_{\text{odd}} \cap Q_{\leq 3} = \{1, 3\}, \tag{1.10}$$

$$P_{\text{odd}} \cup Q_{\leq 3} = \{1, 2, 3, 5\}. \tag{1.11}$$

So "odd and less than or equal to three" is represented by intersection, while "odd or less than or equal to three" is represented by union. If two subsets do not overlap, the classical statement "P and Q" is simply false.

This is the logic of set theory. It is not an extra layer placed on top of classical mechanics; it comes from the fact that a classical state space is a set. The lecture then makes the decisive claim: quantum mechanics does not inherit this logic in the same way. The space of quantum states is not a set of classical alternatives. Only in certain classical limits does it approximately reduce to that kind of picture. The lecture even tells us the replacement in advance: the quantum state space is a vector space.

Before explaining that abstractly, however, Susskind turns to the simplest experiment.

1.3 The Qubit, Its Three Notations, and the Apparatus

The simplest possible quantum system is the quantum analogue of the coin. Susskind calls it a *qubit*. The corresponding classical object would be a *c-bit*, a classical bit, nothing more than an abstract two-state system.

At first the qubit is labeled by the familiar language of heads and tails, but the lecture immediately introduces a more useful variable σ:

$$\sigma \in \{+1, -1\}. \tag{1.12}$$

The three notational systems are then placed in parallel:

$$H \leftrightarrow \sigma = +1 \leftrightarrow \uparrow, \qquad T \leftrightarrow \sigma = -1 \leftrightarrow \downarrow . \tag{1.13}$$

The lecture is careful here. At this stage the arrows are only notation. They are not yet meant to imply that the qubit is literally an ordinary vector in three-dimensional space.

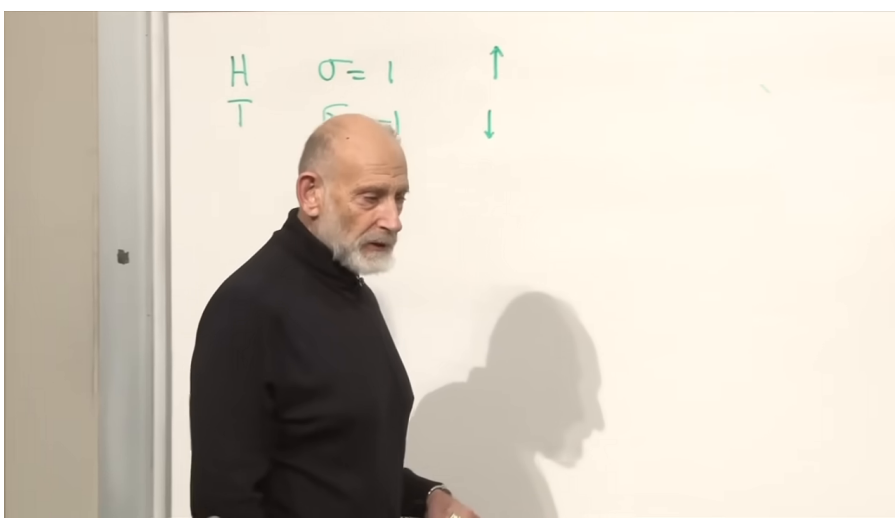

Figure 1.1: Qubit labels $\sigma = \pm 1$ and up/down notation.

One audience member asks what heads and tails have to do with the actual physics. The answer is: nothing. They are only a familiar way to emphasize that there are two alternatives. The variable σ is the mathematically more useful label.

Now the experiment enters. A quantum experiment involves not only the system but also an *apparatus*, which the lecture denotes by A. Unlike the qubit, the apparatus is treated for now as a black box. It has an orientation, a sensor, and a window in which a number can appear. In its

neutral state the window is blank. When the apparatus interacts with the qubit, a faithful detector records one of only two outcomes:

$$\text{readout} \in \{+1, -1\}. \tag{1.14}$$

This is the operational starting point. We are not yet concerned with the detailed inner workings of the detector, only with the fact that such detectors exist and can record the state of the qubit.

The lecture then insists on a crucial point. If the first measurement gives $+1$, and we reset the apparatus and immediately repeat the measurement in the same orientation, we get $+1$ again. Repeating the same measurement confirms the first one:

$$+1 \to +1 \to +1 \to \cdots, \qquad -1 \to -1 \to -1 \to \cdots. \tag{1.15}$$

Thus the measurement is not merely a readout. It is also a preparation. Once the qubit has been found in a given state, repeating the same experiment leaves that result fixed.

This is the first genuinely quantum lesson of the lecture, but it is phrased operationally rather than interpretively. The lecture does not yet speak in terms of wave-function collapse. It speaks in terms of what the apparatus does: the first measurement produces a result; the second, if done immediately in the same orientation, confirms it.

Now the apparatus is turned upside down. A qubit prepared in the state $\sigma = +1$, or $\uparrow$, is measured again. The readout changes sign. Turn the detector upright again and the sign flips back. This already tells us something new. The up/down arrows were introduced as notation, but the experiment shows that there is genuine spatial directionality in the problem. That suggestion is exactly what the next step will sharpen and then partly destroy.

1.4 Detector Rotations and the Breakdown of the Classical Picture

Once the detector has an orientation, a classical guess presents itself immediately. Perhaps the qubit is really something like an ordinary pointer in space, and the apparatus simply measures its component along the detector axis. With the detector upright, the measured component is $+1$. Turn the detector by $180°$, and the component changes sign to -1. So far, the classical picture seems to fit.

But the lecture is setting a trap for that classical intuition. The qubit is prepared "up," and then the detector is turned sideways by $90°$. If the qubit were an ordinary pointer, the horizontal component of a vertical vector would be zero. Classically we would expect

$$\text{horizontal component of an upward pointer} = 0. \tag{1.16}$$

What the apparatus actually returns is not 0. It returns, once again, only $+1$ or -1.

1.4.1 Question & Answer

Question. If the qubit has been prepared "up," why does a sideways detector not read 0, the way a classical detector would read the horizontal component of an ordinary upward pointer?

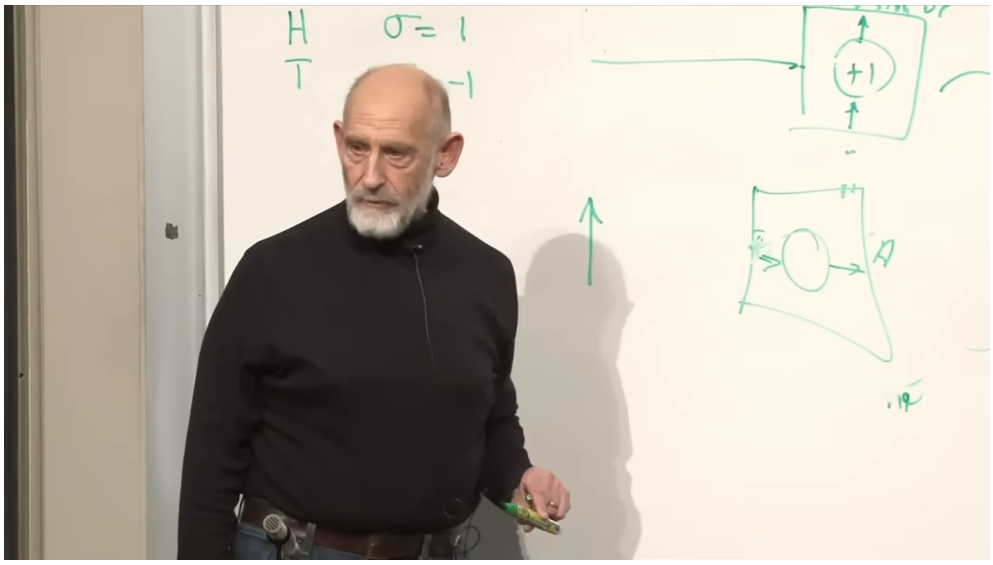

Figure 1.2: Prepared upward state together with upright and sideways analyzer sketches.

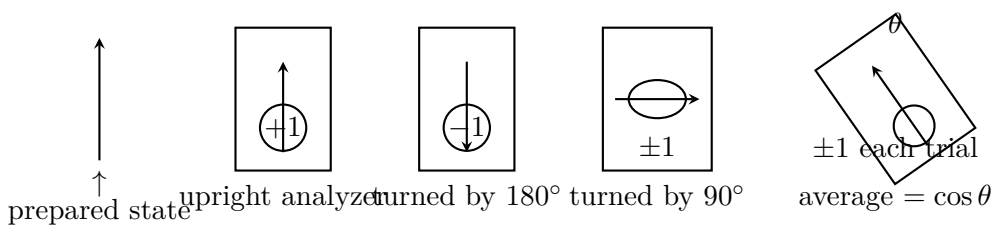

Figure 1.3: Cleaned reconstruction of the detector-orientation experiments. The redraw follows the transcript's logic rather than every literal chalk arrow.

Answer. Because the qubit is not behaving like an ordinary classical pointer. The sideways measurement still yields only the discrete outcomes $+1$ and -1. The first sideways result is random. If it happens to be $+1$, then a repeated sideways measurement confirms that $+1$ again and again. But if we now rotate the detector back to the original upright orientation, the result becomes random once more. The measurement has prepared the qubit relative to the new axis, not revealed a pre-existing classical component.

This is the main conceptual shock of the lecture. Classical component language works for averages, but it fails at the level of individual experiments. A classical upward pointer has zero horizontal component. A quantum-mechanical "up" qubit does not yield 0 under a sideways measurement, ever. It yields only $+1$ or -1.

1.5 Ensembles, Rotated Detectors, and the $\cos\theta$ Law

At this point the lecture makes a distinction that must not be blurred. There is a difference between confirming a measurement on one already measured qubit and determining statistics across many identically prepared qubits.

For the sideways analyzer, a single run gives either $+1$ or -1. If we repeat the same sideways

measurement on that same qubit, we only confirm the already established result. To see the underlying law, we must prepare many qubits in the same initial state and measure each one once.

For the 90° case, the lecture's statement is:

$$\text{each trial gives } +1 \text{ or } -1, \qquad \text{but the ensemble average is 0.} \tag{1.17}$$

This is not because the apparatus is secretly returning intermediate values. It never does. The average is zero only because, over many identically prepared trials, there are as many $+1$'s as -1's.

If we now rotate the detector by an arbitrary angle θ relative to the preparation axis, the same pattern survives. Each single trial still gives only $+1$ or -1, but the average now depends smoothly on θ. As a convenient editorial shorthand, let

$$\overline{\sigma}(\theta) \tag{1.18}$$

denote the ensemble average of the readout for an analyzer tilted by θ. Then the lecture's empirical rule is

$$\overline{\sigma}(0) = 1, \tag{1.19}$$

$$\overline{\sigma}\left(\frac{\pi}{2}\right) = 0, \tag{1.20}$$

$$\overline{\sigma}(\pi) = -1, \tag{1.21}$$

$$\overline{\sigma}(\theta) = \cos\theta. \tag{1.22}$$

The lecture does *not* yet derive detailed probabilities for $+1$ and -1; it gives only the average. That restraint should be preserved here. The point of the experiment is already striking enough: the average behaves as though one were taking the component of a classical vector, but the individual measurements never do.

In fact the lecture emphasizes the two sides together. If the detector is nearly upright, then mostly we see $+1$, with an occasional -1. If it is nearly upside down, then mostly we see -1, with an occasional $+1$. At no stage does the detector return a continuum of values. The continuum appears only after averaging.

1.5.1 Question & Answer

Question. Why do we need a collection of identically prepared qubits? If nothing happens between measurements, why can we not keep using the same qubit and recover the average from repeated measurements on it?

Answer. Because repeated measurements with the same detector do not generate a distribution. They only confirm the outcome of the first measurement. Once a tilted analyzer has produced $+1$ or -1, the qubit has been prepared relative to that tilted axis, and repeating the same tilted measurement merely reproduces the same answer. To learn the statistical law, we need many qubits prepared in the same initial state and measured once each.

The lecture later answers a related question about the sequence of outcomes. For the purposes of this first discussion, the trials are taken to be random and uncorrelated from one prepared qubit to the next. The interesting structure is in the mean, not in higher-order correlations.

This is also where Susskind explicitly postpones the more interpretive language. One can say, if one likes, that the measurement has collapsed the state, but the lecture refuses to lean on that vocabulary yet. Operationally, it is enough to say that the first measurement changes the system, and the second measurement, if carried out immediately in the same orientation, merely keeps what the first one has done.

The same kind of postponement applies to the apparatus itself. Strictly speaking, the detector is also a physical system and ought eventually to be treated quantum mechanically. The lecture acknowledges this point but sets it aside. For now, the world is provisionally divided into detectors and systems. The complication of combining them into a single quantum description is deferred to a later stage.

The lecture now leaves us with the puzzle in its clearest form. The average behavior looks vector-like. The single-shot outcomes do not. That is exactly the moment when the mathematics of vector spaces is brought onto the stage.

1.6 From Pointers to Complex Vector Spaces

Susskind begins the mathematical interlude by separating two meanings of the word *vector*. An ordinary arrow in physical space is something we can picture geometrically. To keep that meaning distinct, he gives it a private name: a *pointer*. The vectors that quantum mechanics needs are more abstract. They are not necessarily arrows in ordinary three-dimensional space at all.

Definition 1.1. A vector space is a collection of mathematical objects with the following basic properties:

1. vectors can be added, and the sum stays in the same space;
2. vectors can be multiplied by scalars, and the result stays in the same space;
3. there is a zero vector;
4. addition is commutative.

The lecture stresses closure under addition and scalar multiplication, and then fills in the zero vector and commutativity when asked. Real numbers provide a real vector space: they are closed under addition and multiplication by real numbers. Complex numbers provide a complex vector space: they are closed under addition and multiplication by complex numbers. The lecture also remarks, briefly, that complex-valued functions of a real variable form a vector space as well. That is an important enlargement of the idea, but here it is only a signpost. The main concrete model of the lecture is simpler.

We consider pairs of complex numbers arranged in a column:

$$|\alpha\rangle \leftrightarrow \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}, \qquad \alpha_1, \alpha_2 \in \mathbb{C}. \tag{1.23}$$

If

$$|\beta\rangle \leftrightarrow \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix},$$

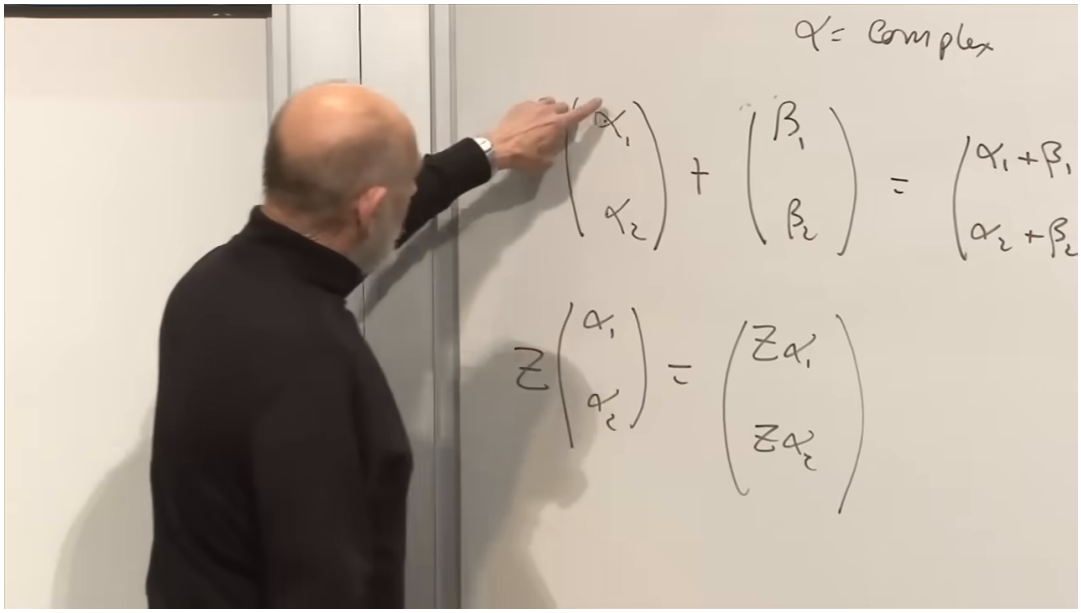

Figure 1.4: Two-component complex column-vector operations.

then the board gives the rules

$$\begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix} + \begin{pmatrix}\beta_1\\ \beta_2\end{pmatrix} = \begin{pmatrix}\alpha_1+\beta_1\\ \alpha_2+\beta_2\end{pmatrix}, \tag{1.24}$$

$$z\begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix} = \begin{pmatrix}z\alpha_1\\ z\alpha_2\end{pmatrix}, \qquad z\in\mathbb{C}. \tag{1.25}$$

This is the concrete realization that the lecture wants us to hold on to. Columns of length two form one vector space. Columns of length three form another. One does not add a two-entry column to a three-entry column, because there is no rule for doing so.

The zero vector is simply

$$|0\rangle = \begin{pmatrix}0\\ 0\end{pmatrix}, \qquad |\alpha\rangle + |0\rangle = |\alpha\rangle, \tag{1.26}$$

and commutativity follows because the entries are ordinary complex numbers:

$$|\alpha\rangle + |\beta\rangle = |\beta\rangle + |\alpha\rangle. \tag{1.27}$$

Now Dirac notation enters more seriously. The abstract vectors are *kets*, written $|A\rangle$. For each ket there is a corresponding element in the dual vector space, written $\langle A|$, and called a *bra*. The correspondence preserves addition and conjugates scalars:

$$\langle A+B| = \langle A| + \langle B|, \tag{1.28}$$

$$\langle zA| = z^*\langle A|. \tag{1.29}$$

For the concrete two-component column,

$$|\alpha\rangle \leftrightarrow \begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix} \qquad \Longrightarrow \qquad \langle\alpha| \leftrightarrow \begin{pmatrix}\alpha_1^* & \alpha_2^*\end{pmatrix}. \tag{1.30}$$

The row form is only a device for keeping track of which space the object lives in, but it is a very convenient one.

With bras and kets in hand, the lecture defines the inner product. If

$$\langle\beta| \leftrightarrow \begin{pmatrix} \beta_1^* & \beta_2^* \end{pmatrix}, \qquad |\alpha\rangle \leftrightarrow \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix},$$

then

$$\langle\beta|\alpha\rangle = \beta_1^*\alpha_1 + \beta_2^*\alpha_2. \tag{1.31}$$

This is the complex analogue of the dot product for pointers: multiply corresponding components and add, except that the bra contributes complex-conjugated components.

Proposition 1.2. *For the two-component complex column model,*

$$\langle A|B\rangle = \langle B|A\rangle^*. \tag{1.32}$$

Moreover,

$$\langle\alpha|\alpha\rangle = |\alpha_1|^2 + |\alpha_2|^2, \tag{1.33}$$

which is real and nonnegative.

Proof. Write

$$\langle\beta|\alpha\rangle = \beta_1^*\alpha_1 + \beta_2^*\alpha_2.$$

Interchanging α and β gives

$$\langle\alpha|\beta\rangle = \alpha_1^*\beta_1 + \alpha_2^*\beta_2,$$

which is the complex conjugate of the previous expression. Hence

$$\langle\beta|\alpha\rangle = \langle\alpha|\beta\rangle^*.$$

Now set $\beta = \alpha$. Then

$$\langle\alpha|\alpha\rangle = \alpha_1^*\alpha_1 + \alpha_2^*\alpha_2 = |\alpha_1|^2 + |\alpha_2|^2.$$

Each term is real and nonnegative, so the whole expression is real and nonnegative. □

The lecture interprets $\langle A|A\rangle$ as the square of the length of the vector. Thus even though the components are complex, the length is real and positive.

Orthogonality is then defined in the expected way.

Definition 1.3. Two vectors are orthogonal if their inner product vanishes:

$$\langle A|B\rangle = 0. \tag{1.34}$$

1.6.1 Question & Answer

Question. Does the order matter? Is $\langle B|A\rangle$ the same as $\langle A|B\rangle$?

Answer. No. In a complex vector space the two expressions are not generally equal; they are complex conjugates of one another:

$$\langle A|B\rangle = \langle B|A\rangle^*.$$

But this is enough to make orthogonality symmetric. If $\langle A|B\rangle = 0$, then $\langle B|A\rangle = 0^* = 0$. So orthogonality does not depend on order, even though the inner product itself does.

Finally the lecture turns to dimension. Rather than taking dimension to mean "number of components" from the outset, it defines it abstractly as the maximal number of mutually orthogonal nonzero vectors in the space. In the concrete two-component model, a simple orthogonal pair is

$$|e_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |e_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad \langle e_1|e_2\rangle = 0. \tag{1.35}$$

Once we have found two mutually orthogonal nonzero vectors in this space, there is no third nonzero vector orthogonal to both. Thus the space is two-dimensional. In this example, the abstract definition agrees with the obvious concrete one: the dimension equals the number of components in the column.

The lecture closes the mathematical interlude by hinting that there is still more structure to come. There will be other products of vectors, including outer products that produce matrices, and the general ideas will later be extended to vector spaces of functions. But the central point of tonight's interlude is already in place: the space of states is a complex vector space, not a set.

1.7 Summary

This first lecture places two blackboards side by side. On one blackboard are the experiments. A qubit yields only the two outcomes $+1$ and -1. A repeated measurement in the same orientation confirms and therefore prepares a state. Turning the analyzer by $180°$ flips the sign. Turning it by $90°$ destroys classical definiteness and gives random ± 1 outcomes with zero mean. Turning it by an arbitrary angle θ produces an ensemble average $\cos\theta$.

On the other blackboard is the mathematics. Classical state spaces are sets, and classical logical operations are set-theoretic operations on subsets. Quantum state spaces are different. They are complex vector spaces, with kets, bras, inner products, norms, orthogonality, and dimension. The lecture does not yet weld these two stories together completely. It prepares them. Next time the abstract vector-space machinery will be brought back to the qubit experiments and made to represent their strange logic directly.

Chapter 2

Spin, Measurement, and the Two-Dimensional State Space

This lecture is presented as a review, but it is really a reorganization of the subject around its central difficulty. Leonard Susskind renames the qubit a spin, cleans up the distinction between ordinary three-vectors and abstract state-vectors, and then deliberately interrupts the march toward operators in order to talk about measurement and logic first. That order matters. The strange logic of quantum measurements is not an afterthought; it is the reason vector spaces enter at all. These notes follow that unfolding closely, with the surviving blackboard record kept visible where it clarifies the sequence of the argument. The lecture record itself is due to Leonard Susskind, with curation by LazyingArt LLC and Video2Book.

2.1 From Qubit To Spin

We begin with a change of language. The simple one-qubit system of the previous lecture will now be called a spin. Susskind also sharpens a distinction that we will need constantly from this point onward.

Definition 2.1. A *three-vector* is a vector in ordinary real three-dimensional space. When we speak simply of a *vector*, unless stated otherwise, we mean an element of an abstract vector space.

This is not cosmetic terminology. The spin does seem to possess some kind of orientation in space, but the object that will ultimately describe its state is not itself an ordinary arrow in space. The lecture is constantly moving between these two meanings of the word *vector*, and it is important not to let them blur together.

That clarification leads directly to the next point: the role of measurement. In classical physics one may idealize measurements as arbitrarily gentle. One can, in principle, find out what one wants to know about a system while disturbing it by an arbitrarily small amount. Quantum mechanics is not like that. Measurements inevitably disturb something. They do not necessarily destroy every fact about the system, but they do prevent us from pretending that a system and the apparatus measuring it can be discussed in complete isolation from one another.

That is why Susskind pauses here. Before returning to vector spaces and operators, he wants to talk first about the logic of the subject. The logic is different because the measurements are different.

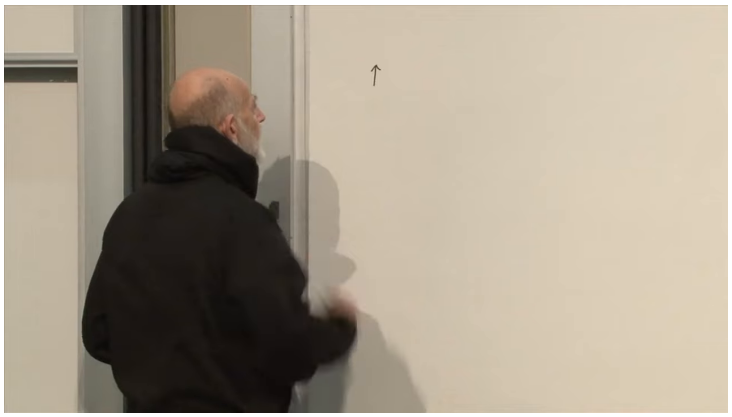

Figure 2.1: Spin introduced as a single upward arrow.

2.2 The Spin And The Apparatus

Let us go back to the simplest operational picture. The spin is drawn as a little arrow, not because we have already proved that it is a classical arrow, but because anything we can draw on a blackboard will be classical-looking.

We may echo the board's notation by writing

$$\uparrow$$

but the picture should be read cautiously. It is only a provisional visualization.

The apparatus is imagined as a device with a window that displays a number and with an orientation that tells us which way it is pointing. The crucial empirical fact is very simple: no matter how the apparatus is oriented, and no matter what has happened to the spin beforehand, the apparatus returns only two possible outcomes,

$$\sigma_{\hat{m}} = \pm 1, \tag{2.1}$$

where $\hat{m}$ denotes the orientation of the apparatus. When the apparatus points upward we say that it measures the z-component of spin, and when it points to the right we say that it measures the x-component. The lecture's convention is

$$z \text{ up}, \qquad x \text{ to the right}, \qquad y \text{ out of the blackboard}.$$

At this stage the word "component" is only a label. Susskind is explicit about this. We are not yet entitled to think of $\sigma_z, \sigma_x, \sigma_y$ as classical components of an ordinary three-vector. All we know is that the apparatus, when pointed in any direction, answers with $+1$ or -1.

A measurement also serves as a preparation. If we point the apparatus upward and obtain $+1$, then we say that the spin has been prepared "up" along the z-axis.

2.2.1 Question & Answer

If measurements disturb systems, why do repeated measurements of the same component still reproduce the same result? Because quantum disturbance is not the statement that every measurement randomizes everything. Susskind's answer is: a measurement affects some

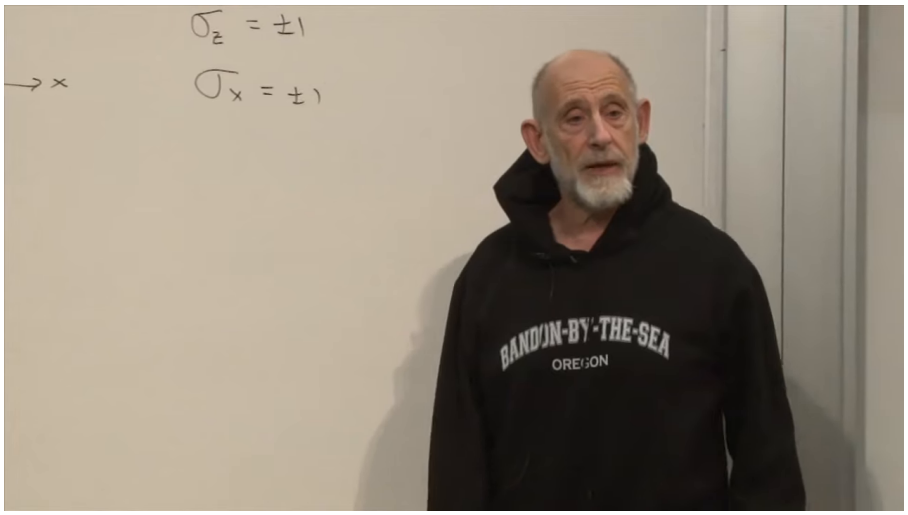

Figure 2.2: Outcome labels for σ_z and σ_x.

$$\longrightarrow x \qquad \begin{array}{l} \sigma_z = \pm 1 \\ \\ \sigma_x = \pm 1 \end{array}$$

Figure 2.3: Cleaned reconstruction of the component labels and the x-direction cue.

things, but not that same thing. If we measure σ_z and obtain $+1$, then measure σ_z again, with the same apparatus orientation and no intervention in between, we get $+1$ again, and again. In that sense quantum mechanics still permits reproducible experiments.

This is not a minor remark. Without such reproducibility there would be no stable notion of preparation. A measurement both reveals and prepares: after the first z-measurement, the spin is definite with respect to the very same question.

2.3 Direction Without Classical Components

Now we test whether this object really has some notion of directionality. If we measure the spin with the apparatus pointing up and obtain $+1$, then turn the apparatus completely over and measure again, we obtain -1. That is not how a thermometer behaves. Turning a thermometer upside down does not make it measure minus the temperature. So already the spin is telling us that orientation matters.

The same is true when the apparatus is rotated sideways. We are then asking for what the lecture calls the x-component of the spin. The board stabilizes the notation as follows.

The upper σ_z-line is slightly cropped in the surviving frame, but the lecture's meaning is clear from the transcript and neighboring board states. A clean reconstruction is

So the spin certainly responds to the orientation of the apparatus. But that does *not* mean it is an ordinary unit vector. If a classical vector of unit length were measured along arbitrary axes, its components would range continuously from -1 to $+1$. Here the apparatus stubbornly refuses to produce anything except the extreme values ± 1. That is the first major tension of the lecture: the

spin behaves enough like something directional to invite the picture of an arrow, but it does not behave like a classical vector when one measures its components.

2.3.1 Question & Answer

Does the little arrow mean that there is genuine geometric direction, or is it only a provisional visualization? It is only a visualization, and Susskind says so repeatedly. The board picture is classical because the board itself is classical. One should not infer too much from the drawing. What the experiments do justify is weaker and more interesting: they show that the spin carries directional information in an operational sense. Rotate the apparatus and the statistics change; turn the apparatus over and the sign flips. That much is real. But the individual measurement outcomes are still only ± 1, and there is no classical geometry yet hiding inside a single event.

2.4 Average Values Recover Vector-Like Behavior

The next step is the key one. Suppose the spin has been prepared so that

$$\sigma_z = +1.$$

Now rotate the apparatus and measure σ_x. The result is no longer fixed. Half the time it comes out $+1$, and half the time it comes out -1. Therefore

$$\langle \sigma_x \rangle = 0. \tag{2.2}$$

The same is true if the apparatus is rotated into the y-direction:

$$\langle \sigma_y \rangle = 0. \tag{2.3}$$

Thus a spin prepared up along z has vanishing average x- and y-components, even though every single run still gives only ± 1. If instead the spin is prepared so that $\sigma_x = +1$, then subsequent y- or z-measurements are likewise split evenly between $+1$ and -1.

Only after these special cases are on the table does Susskind generalize. Prepare the spin along a unit direction $\hat{n}$, then measure along another unit direction $\hat{m}$. Every individual run still yields only ± 1, but the average value is

$$\langle \sigma_{\hat{m}} \rangle = \hat{n} \cdot \hat{m} = \cos \theta_{nm}. \tag{2.4}$$

On the board, the formula appears in the shorter lecture notation

$$\langle \sigma_m \rangle = n \cdot m. \tag{2.5}$$

The surviving frame is strong here. The first handwritten symbol could be misread in isolation, but the transcript clearly supports $n \cdot m$, and that is the form we should preserve.

There is a small terminological bump in the lecture at this point. Susskind insists that the quantity in angle brackets is an ordinary average over many runs, even though physicists also often call it an expectation value. The mathematical content is the standard one: repeat the experiment many times and take the mean.

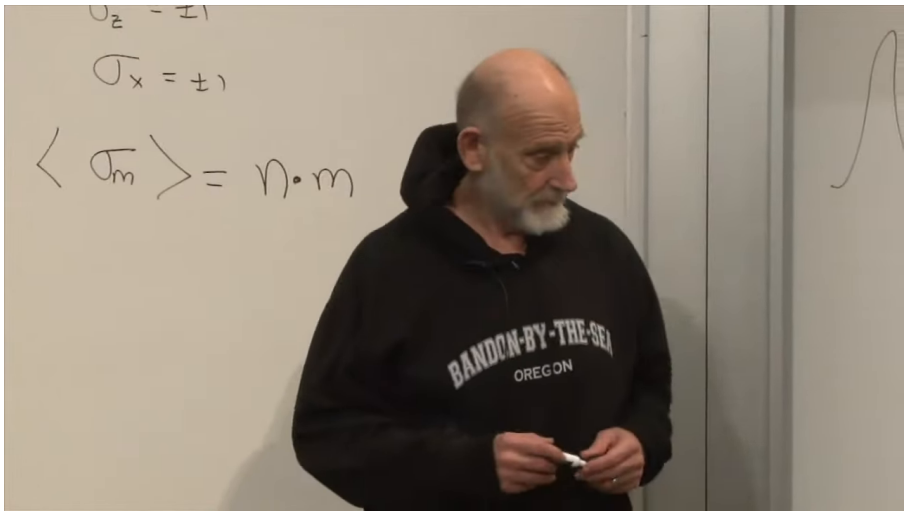

Figure 2.4: Average spin component written as $\langle \sigma_m \rangle = n \cdot m$.

The formula becomes more concrete if we write the probabilities explicitly. Let

$$p_+ = P(\sigma_{\hat{m}} = +1), \qquad p_- = P(\sigma_{\hat{m}} = -1).$$

Then

$$p_+ + p_- = 1, \tag{2.6}$$
$$p_+ - p_- = \langle \sigma_{\hat{m}} \rangle = \hat{n} \cdot \hat{m}. \tag{2.7}$$

Solving these two equations gives

$$p_+ = \frac{1 + \hat{n} \cdot \hat{m}}{2}, \qquad p_- = \frac{1 - \hat{n} \cdot \hat{m}}{2}. \tag{2.8}$$

So the apparatus never learns how to output a projection directly. It remains a ± 1 machine. The classical-looking projection emerges only after averaging.

There is an important limiting case that the lecture draws attention to. If $\hat{n} = \hat{m}$, then $\hat{n} \cdot \hat{m} = 1$. Since the outcomes can only be ± 1, an average of $+1$ means every single run gave $+1$. Likewise, if $\hat{n} = -\hat{m}$, every run gives -1. The average law therefore contains the reproducibility statements as special cases.

2.5 Quantum Logic Through Propositions

At this point the lecture makes its most deliberate pivot. Before more vector spaces and operators, we are asked to think about logic. Classical logic can be modeled in terms of sets: the states of a classical system form a set, and a proposition corresponds to the subset of states for which the proposition is true.

Thus classical negation, conjunction, and inclusive disjunction are represented by

$$\neg A = A^c, \qquad A \wedge B = A \cap B, \qquad A \vee B = A \cup B. \tag{2.9}$$

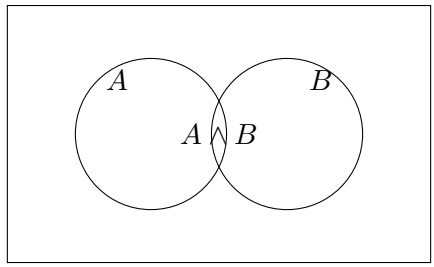

Figure 2.5: Classical propositions as subsets of a state-space.

Susskind pauses to emphasize that in logic the symbol $\vee$ means the inclusive OR, not the everyday exclusive OR. If A is true, or B is true, or both are true, then $A \vee B$ is true.

Now let us transport this language to the spin. Take the two propositions

$$A : \sigma_z = +1, \qquad B : \sigma_x = +1. \tag{2.10}$$

Each can be checked individually without ambiguity. To check A, we measure along the z-axis. To check B, we measure along the x-axis. The trouble begins only when we ask how to verify the compound proposition $A \vee B$.

2.5.1 Question & Answer

How can $A \vee B$ become order-dependent in quantum mechanics? To make the point as sharply as possible, Susskind imagines a "ghost in the woodwork" who has secretly prepared the spin with $\sigma_z = +1$. We ourselves do not know this, but the hidden preparation lets us compare two verification procedures for the same compound statement.

If we test A first, then the very first measurement returns $+1$. We are finished. Since A is true, the proposition $A \vee B$ is certainly true.

Now reverse the order. Test B first. Since the spin was secretly prepared with $\sigma_z = +1$, the first x-measurement gives $+1$ or -1 with equal probability:

$$P(B \text{ true first}) = \frac{1}{2}, \qquad P(B \text{ false first}) = \frac{1}{2}.$$

But a measurement of B does not merely reveal an answer; it prepares the spin along the x-axis. So if the first result was not $+1$, we must then check A on a spin that is now definite with respect to x, not z. The subsequent z-measurement is again random:

$$P(A \text{ true after testing } B) = \frac{1}{2}, \qquad P(A \text{ false after testing } B) = \frac{1}{2}.$$

Therefore the probability that both tests fail is

$$P(\neg B \text{ and then } \neg A) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}. \tag{2.11}$$

So in one order, $A \vee B$ is certainly true; in the reversed order, it has a 25% chance to come out false. That is the phenomenon the lecture wants us to see. In classical logic, $A \vee B$ is symmetric

in A and B. In quantum mechanics, any operational meaning given to the verification of $A \vee B$ depends on the order in which the tests are carried out.

This is not merely the bland statement that "measurements disturb systems." The sharper statement is that compound propositions inherit that disturbance and cease to have the classical symmetry one naively expects. Susskind allows the more technical summary that propositions correspond to projection operators, and projection operators need not commute:

$$P_A P_B \neq P_B P_A. \tag{2.12}$$

That is not yet a full operator theory, but it is the right mathematical slogan for the logical obstruction.

2.6 Why Vector Spaces Enter

Only after the logical obstruction has been exposed does the lecture return to vector spaces. This timing is essential. Linear algebra is not being pasted on top of the physics as an abstract formal language; it is being introduced because the distorted quantum logic lives naturally inside the structure of a complex vector space.

The essential ingredients are the familiar ones. Vectors can be added. They can be multiplied by complex numbers. Every ket-vector has a corresponding bra-vector, and a bra may be paired with a ket to produce an inner product. If $|A\rangle$ and $|B\rangle$ are state-vectors, then

$$\langle A|B\rangle = \langle B|A\rangle^*. \tag{2.13}$$

The square of the length of a vector is

$$\langle A|A\rangle, \tag{2.14}$$

and this quantity is real and nonnegative. Two vectors are orthogonal when

$$\langle A|B\rangle = 0. \tag{2.15}$$

Dimension is then defined in the way Susskind likes to emphasize: it is the maximum number of mutually orthogonal vectors one can find in the space. In a two-dimensional complex vector space one can find two mutually orthogonal vectors, but no third one independent of them.

For the simple spin, the postulate is that the state-space is precisely such a space:

$$\mathcal{H}_{\text{spin}} \cong \mathbb{C}^2. \tag{2.16}$$

That postulate summarizes a great deal. It says that although the apparatus may be pointed in continuously many directions, the abstract state-space of the single spin is nevertheless two-dimensional. The different prepared states are not separate dimensions; they are different vectors in the same two-dimensional complex space.

2.7 Constructing The Spin State Space

We now build the state-space explicitly, in the same heuristic order as the lecture itself. The style here is intentionally provisional. Susskind is not pretending to derive everything from a finished formalism; he is guessing, checking, and refining the vectors so that they reproduce the measurement facts.

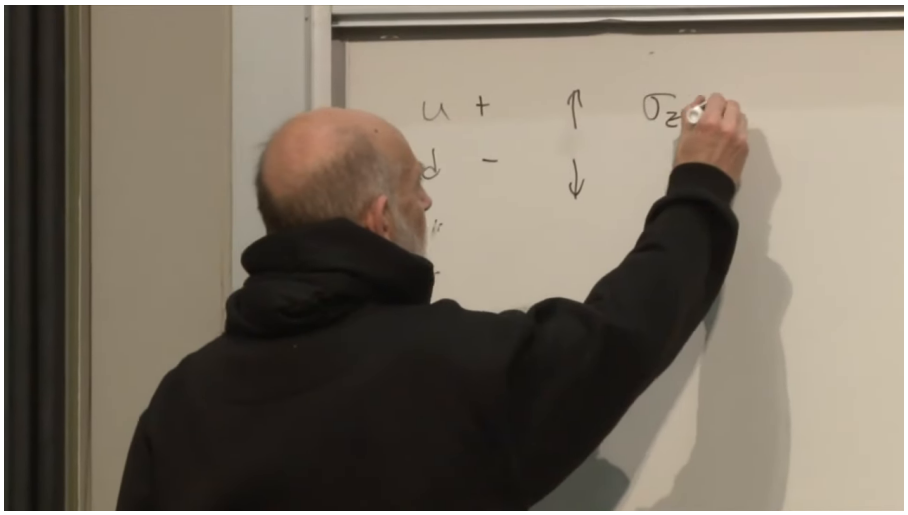

Figure 2.6: Up/down notation translated into σ_z-language.

2.7.1 The z-basis

We begin with the states prepared by an apparatus aligned with the z-axis. The lecture insists that the same state may be written in several different "languages": as a word, as a sign, as a little arrow, and as a σ-statement.

A clean reconstruction of the row-wise board dictionary is

$$u \quad + \quad \uparrow \quad \sigma_z = +1 \quad |u\rangle, \tag{2.17}$$

$$d \quad - \quad \downarrow \quad \sigma_z = -1 \quad |d\rangle. \tag{2.18}$$

The completion $\sigma_z = +1$ in the first row is partly obscured in the surviving frame, but it is clearly supported by the spoken lecture and by the later board state.

These two states are not negatives of one another. They are orthogonal:

$$\langle u|d\rangle = 0. \tag{2.19}$$

The physical reason is the lecture's operational criterion: if two states can be distinguished unambiguously by an experiment, then they are orthogonal. A z-measurement distinguishes $|u\rangle$ from $|d\rangle$ with certainty.

Since the space is assumed to be two-dimensional, any state can be expanded in this basis:

$$|A\rangle = \alpha_u|u\rangle + \alpha_d|d\rangle, \tag{2.20}$$

with $\alpha_u, \alpha_d \in \mathbb{C}$.

The transcript is slightly garbled when Susskind states the probability rule, but the intended content is standard and clear:

$$P(u) = |\alpha_u|^2, \qquad P(d) = |\alpha_d|^2. \tag{2.21}$$

Because probabilities must add to one,

$$|\alpha_u|^2 + |\alpha_d|^2 = 1. \tag{2.22}$$

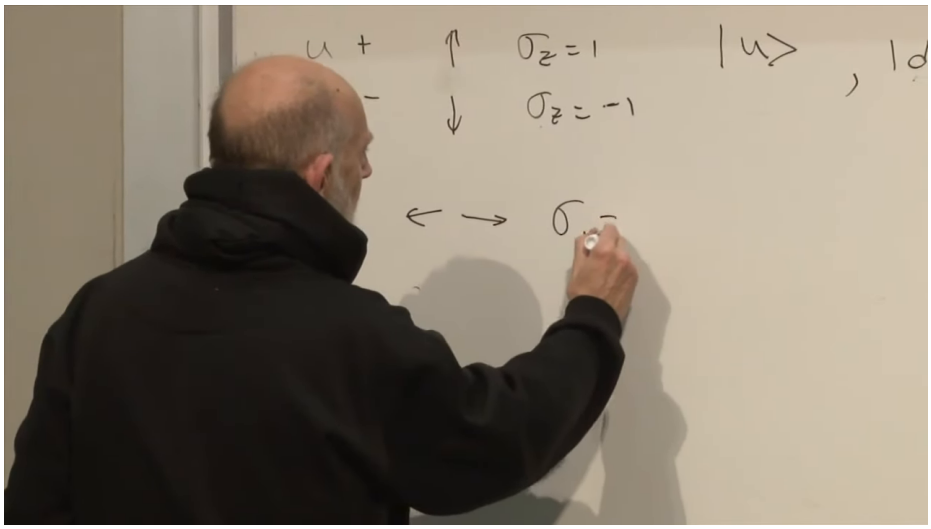

Figure 2.7: The z-basis dictionary extended toward the x-basis states.

This is exactly the normalization condition $\langle A|A\rangle = 1$, as one sees by using the orthonormality of $|u\rangle$ and $|d\rangle$:

$$\langle A|A\rangle = (\alpha_u^*\langle u| + \alpha_d^*\langle d|)(\alpha_u|u\rangle + \alpha_d|d\rangle) \tag{2.23}$$

$$= |\alpha_u|^2\langle u|u\rangle + \alpha_u^*\alpha_d\langle u|d\rangle + \alpha_d^*\alpha_u\langle d|u\rangle + |\alpha_d|^2\langle d|d\rangle \tag{2.24}$$

$$= |\alpha_u|^2 + |\alpha_d|^2. \tag{2.25}$$

Thus physical states correspond to normalized vectors.

2.7.2 Question & Answer

In what sense are $|u\rangle$ and $|d\rangle$ orthogonal if the pictures "up" and "down" lie on one line in ordinary space? Because orthogonality here is not ordinary perpendicularity in real space. The lecture eventually sharpens this point: in the space of states, orthogonality means physical distinguishability. $|u\rangle$ and $|d\rangle$ are orthogonal because a single well-designed experiment can tell them apart without ambiguity. By contrast, $|u\rangle$ and $|r\rangle$ are not orthogonal: a z-measurement on $|r\rangle$ can still return "up" with probability 1/2. In that sense they overlap.

The safest rule is therefore this: do not confuse orthogonality in state-space with orthogonality of arrows in three-dimensional space.

2.7.3 The x-basis: right and left

Now rotate the apparatus into the x-direction. This gives two new states, right and left, corresponding to $\sigma_x = +1$ and $\sigma_x = -1$.

The board layout is important here: the z-basis rows remain at the top while a second, lower row begins for the left-right states. A clean reconstruction is

Let us denote the corresponding states by

$$|r\rangle \leftrightarrow \sigma_x = +1, \qquad |l\rangle \leftrightarrow \sigma_x = -1.$$

$$\begin{array}{cccc} u & + & \uparrow & \sigma_z = +1 \\ d & - & \downarrow & \sigma_z = -1 \\ & & \longrightarrow \longleftarrow & \sigma_x = \pm 1 \end{array}$$

Figure 2.8: Cleaned reconstruction of the board as the x-basis is introduced.

Now comes the lecture's characteristic move: we know $|r\rangle$ must be some linear combination of $|u\rangle$ and $|d\rangle$, so let us write

$$|r\rangle = \alpha_u|u\rangle + \alpha_d|d\rangle.$$

What do we know about the coefficients? If the spin is prepared in the right state and we then measure up or down, each occurs with probability 1/2. Therefore

$$|\alpha_u|^2 = \frac{1}{2}, \qquad |\alpha_d|^2 = \frac{1}{2}.$$

The simplest guess is

$$|r\rangle = \frac{1}{\sqrt{2}}(|u\rangle + |d\rangle). \tag{2.26}$$

What about $|l\rangle$? Right and left are unambiguously distinguishable by a single x-measurement, so they must be orthogonal. The natural candidate is

$$|l\rangle = \frac{1}{\sqrt{2}}(|u\rangle - |d\rangle). \tag{2.27}$$

Indeed,

$$\langle l|r\rangle = \frac{1}{2}(\langle u| - \langle d|)(|u\rangle + |d\rangle) = \frac{1}{2}(1-1) = 0. \tag{2.28}$$

Thus the x-basis is

$$|r\rangle = \frac{1}{\sqrt{2}}(|u\rangle + |d\rangle), \tag{2.29}$$

$$|l\rangle = \frac{1}{\sqrt{2}}(|u\rangle - |d\rangle). \tag{2.30}$$

Adding and subtracting these relations gives the inverse formulas

$$|u\rangle = \frac{1}{\sqrt{2}}(|r\rangle + |l\rangle), \tag{2.31}$$

$$|d\rangle = \frac{1}{\sqrt{2}}(|r\rangle - |l\rangle). \tag{2.32}$$

This reciprocity is one of the points the lecture emphasizes: had we started our experiments with the apparatus aligned along x, we might have taken $|r\rangle$ and $|l\rangle$ as basic and then discovered that $|u\rangle$ and $|d\rangle$ were their linear combinations.

2.7.4 The y-basis: in and out

Finally the apparatus is turned along the y-axis, giving the states "in" and "out" of the blackboard:

$$|i\rangle \leftrightarrow \sigma_y = +1, \qquad |o\rangle \leftrightarrow \sigma_y = -1.$$

Here the notation deserves one warning. The ket $|i\rangle$ denotes the in-state, while the plain symbol i remains the imaginary unit. The lecture itself briefly stumbles over this collision of notation.

Again we proceed heuristically. Equal up/down probabilities require equal moduli of the coefficients, and orthogonality requires the pair to be mutually orthogonal. The lecture states the answer:

$$|i\rangle = \frac{1}{\sqrt{2}}(|u\rangle + i|d\rangle), \tag{2.33}$$

$$|o\rangle = \frac{1}{\sqrt{2}}(|u\rangle - i|d\rangle). \tag{2.34}$$

The orthogonality check is straightforward. In the (u, d)-basis,

$$|i\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}, \qquad |o\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix},$$

so

$$\langle i|o\rangle = \frac{1}{2}\left(1 + (-i)(-i)\right) = \frac{1}{2}(1-1) = 0. \tag{2.35}$$

The lecture then worries, a little theatrically, about whether these states are also symmetrically related to the x-basis. One can verify this neatly by rewriting $|i\rangle$ in terms of $|r\rangle$ and $|l\rangle$:

$$|i\rangle = \frac{1}{\sqrt{2}}\left(\frac{|r\rangle + |l\rangle}{\sqrt{2}} + i\frac{|r\rangle - |l\rangle}{\sqrt{2}}\right) \tag{2.36}$$

$$= \frac{1+i}{2}|r\rangle + \frac{1-i}{2}|l\rangle. \tag{2.37}$$

Now

$$\left|\frac{1+i}{2}\right|^2 = \left|\frac{1-i}{2}\right|^2 = \frac{1}{2}, \tag{2.38}$$

so a spin prepared in the in-state has probability 1/2 to be found right and probability 1/2 to be found left. The same holds for $|o\rangle$. This is the symmetry the lecture was aiming at when it said that the three basis pairs are "nicely symmetrically related."

The standard column-vector representatives in the (u, d)-basis are therefore

$$|u\rangle = \begin{pmatrix}1\\ 0\end{pmatrix}, \qquad |d\rangle = \begin{pmatrix}0\\ 1\end{pmatrix}, \tag{2.39}$$

$$|r\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ 1\end{pmatrix}, \qquad |l\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -1\end{pmatrix}, \tag{2.40}$$

$$|i\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}, \qquad |o\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}. \tag{2.41}$$

The lecture then asks whether there could be a fourth pair, as symmetrically related to the first three as they are to one another. The answer is no. That absence is deeply tied to the fact that ordinary space has three dimensions. The striking fact is that the continuously many spatial directions of a spin are being represented inside a two-dimensional complex state-space.

2.7.5 Phase Indifference And Parameter Count

How much information in the pair (α_u, α_d) is physically meaningful? Two complex amplitudes contain four real parameters, but two of them are redundant.

First, normalization removes one parameter:

$$|\alpha_u|^2 + |\alpha_d|^2 = 1. \tag{2.42}$$

Second, the overall phase has no physical effect. If we multiply the whole state by a common phase factor,

$$|A\rangle \longmapsto e^{i\theta}|A\rangle, \tag{2.43}$$

then all measurable probabilities remain unchanged. Thus the counting is

$$4 \text{ real parameters} - 1 \text{ normalization} - 1 \text{ overall phase} = 2 \text{ real parameters}. \tag{2.44}$$

This is the lecture's closing conceptual bridge. A direction in ordinary three-dimensional space is also specified by two real numbers, for example two angles. The agreement is not accidental. It is a first glimpse of the deep relation between a normalized two-component spin state and a direction in real space.

At the same time, Susskind is careful not to overclaim. We are not yet deriving that relation rigorously. We are finding vectors that fit the experiments, and discovering that they fit astonishingly well.

2.8 Summary

The lecture has unfolded in a very definite order. We began with operational facts: an oriented apparatus measures only $+1$ or -1, repeated measurements of the same component are reproducible, and rotating the apparatus changes the statistics in a way that shows genuine directional structure. We then saw the central tension: the spin behaves directionally, but not by having classical components event by event. Classical vector behavior reappears only in averages, through

$$\langle \sigma_{\hat{m}} \rangle = \hat{n} \cdot \hat{m}.$$

From there the lecture pivoted to logic. The compound proposition $A \vee B$, with $A : \sigma_z = +1$ and $B : \sigma_x = +1$, does not behave classically under verification. Its operational meaning depends on the order of testing. That logical distortion is what leads us to vector spaces.

Finally, the lecture built the two-dimensional complex state-space of a single spin. The z-basis states $|u\rangle, |d\rangle$, the x-basis states $|r\rangle, |l\rangle$, and the y-basis states $|i\rangle, |o\rangle$ were constructed as a coherent family of normalized vectors reproducing the measurement facts. The picture is already powerful, but it is still heuristic. The lecture ends by saying so plainly: we have been guessing, checking, and testing. The next step is to make rigorous the relation between these state-vectors and the observables $\sigma_x, \sigma_y, \sigma_z$.